

Zachary Komarnisky

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PROFESSIONAL SUMMARY

Digital Agriculture researcher and developer building open-source AI and data aggregation tools for Canadian precision agriculture. Developing RETA_ML, which is a graph neural network system for spatial outlier detection in field sensor data and CropCore, a multi-source data pipeline that aggregates satellite, weather, soil, and elevation data into actionable farm-level insights. Experienced working across the full agricultural data pipeline from field data collection to spatial analysis and ML-driven decision support.

TECHNICAL SKILLS

AI & Machine Learning

- Python (PyTorch, PyG), Graph Neural Networks (HeteroGAT), XGBoost
- Optuna hyperparameter optimization, focal loss, temperature scaling/calibration
- Scikit-learn, Pandas, NumPy

Data Aggregation & Pipelines

- Multi-source data integration (satellite, weather APIs, soil, elevation)
- GeoPandas, pyproj, Shapely, spatial analysis
- REST APIs (Open-Meteo, Copernicus CDSE, AWS S3), concurrent processing (ThreadPoolExecutor)
- SQLite, GeoPackage, Parquet, Shapefile, GeoTIFF

GIS & Remote Sensing

- QGIS (plugin development), ArcGIS, Sentinel-2 imagery
- NDVI, NDWI, False Color composites, DEM processing
- Drone/UAV data processing, Pix4D, Emlid, Trimble Catalyst

Tools & Platforms

- Git/GitHub, GitHub Actions CI/CD
- Linux (Ubuntu), Jupyter Notebook, PyCharm
- PySide6, FastAPI, CustomTkinter

PROJECTS — OPEN SOURCE

CropCore github.com/DOZER3530-Fishing/CropCore

- Desktop analytics pipeline aggregating Sentinel-2 NDVI, Open-Meteo weather, CDEM/SRTM elevation, and Alberta Open Data soil into per-field and farm-level reports
- Produces PDF, HTML, CSV, and QGIS-compatible outputs; NDVI time-lapse animations and optional SMTP email delivery
- Field stress index (0–100 scale), review priority ranking, and linear regression yield forecasting (wheat, barley, canola) with drought stress penalties

RETA_ML (Private, continuing research)

- GNN-based outlier detection for spatial agricultural sensor data; 4-class classification (Clean, Operational, Global Outlier, Local Outlier)
- HeteroGAT model with temporal, spatial (BallTree), and transect graph edges; pure Python/GeoPandas preprocessing, no QGIS dependency
- Masked hierarchical focal loss, temperature scaling calibration, Optuna hyperparameter sweep; evaluated against RETA threshold baseline and XGBoost

RETA — Robust Error-treatment Toolkit for Agriculture github.com/Dozer3530/RETA

- Unified algorithmic framework for cleaning yield monitor and spatial agricultural data via 3-phase sequential pipeline (Operational → Global → Local errors)
- CRS-aware using pyproj; deployable as QGIS plugin or headless Python script

Lapwing github.com/Dozer3530/Lapwing

- Sentinel-2 L2A satellite imagery downloader for agricultural field boundaries; all-time, cloud-filtered retrieval via Copernicus CDSE

- Outputs RGB, NDVI, NDWI, False Color, and DEM as analysis-ready GeoTIFFs clipped to field boundary

Magpie github.com/Dozer3530/Magpie

- Weekly crop-monitoring package builder merging Survey123 field observations and PT2R lab nutrient data into Excel and GeoPackage deliverables
- Dual frontend: PySide6 desktop + FastAPI web server on a shared service layer; SQLite storage

Perch github.com/Dozer3530/Perch

- Windows GUI for sorting multispectral MicaSense drone imagery into band-specific folders; auto-detects camera from EXIF, parallel processing, PyInstaller single-file distribution

Plover github.com/Dozer3530/Plover

- QGIS plugin solving the Traveling Salesperson Problem for field routing. Built for soil sampling and sensor verification on research plots

WORK EXPERIENCE

Olds College Centre for Innovation

Olds, AB

Research Assistant

Summer 2025-2026

- Collected and processed field data using drones, soil moisture probes, and GPS equipment across multiple applied research projects with senior researchers and industry partners
- Provided technical support and troubleshooting for precision ag sensors and equipment in the field
- Supported data-driven research workflows contributing to agricultural productivity and sustainability outcomes

Town of Carstairs — Operational Services

Carstairs, AB

Seasonal Laborer

2022–2024

- Operated landscaping and maintenance equipment; facility and infrastructure upkeep

EDUCATION

Olds College of Agriculture and Technology

Olds, AB

Bachelor of Digital Agriculture

Expected 2027

Co-Curricular Record: Student Research Experience, 2026

CERTIFICATIONS

- Small RPAS Pilot Certificate — Basic Operations (VLOS), Transport Canada, 2025 (PC2516275416 / TC2516228123)

REFERENCES

Available upon request